

Day 4 : Flutter Layout Widgets

1. Introduction to Flutter Layout Widgets

Flutter provides powerful layout widgets that help developers arrange UI elements efficiently. Layout widgets define how child widgets are displayed on the screen and adapt to different device sizes. Understanding layouts is essential for building responsive and visually appealing mobile applications.

Flutter follows a widget-based architecture where every UI component is a widget. Layout widgets help organize these components into rows, columns, grids, lists, and layered structures.

2. Row and Column Widgets

Row Widget

The Row widget arranges its children horizontally from left to right.

Important Properties:

- `mainAxisAlignment` – Controls horizontal alignment.
- `crossAxisAlignment` – Controls vertical alignment.
- `mainAxisSize` – Determines how much horizontal space the row occupies.

Common Values of `mainAxisAlignment`:

- `start`
- `end`
- `center`
- `spaceBetween`
- `spaceAround`
- `spaceEvenly`

Column Widget

The Column widget arranges its children vertically from top to bottom.

Uses:

- Forms
- Profile pages
- Dashboard sections
- Settings screens

Nested Rows and Columns

Rows and Columns can be combined to create complex layouts. For example, a profile card may contain a Row for the image and user information, while the information section contains a Column for name, email, and phone number.

3. Flexible Layouts

Expanded Widget

The Expanded widget forces a child widget to occupy the remaining available space inside a Row, Column, or Flex.

Advantages:

- Creates responsive layouts
- Prevents overflow issues
- Distributes space evenly

Flexible Widget

Flexible allows a widget to take available space without forcing it to fill all remaining space.

Difference Between Expanded and Flexible

Expanded	Flexible
Occupies all available space	Uses only required space
More rigid	More flexible
Good for equal distribution	Good for adaptive layouts

Flex Factor

The flex property controls how much space each widget receives relative to others.

Example:

- Widget A (flex: 1)
- Widget B (flex: 2)

Widget B receives twice as much space as Widget A.

4. Stack Widget

The Stack widget places widgets on top of each other, similar to layers.

Positioned Widget

Used inside a Stack to place widgets at specific positions.

Alignment

Controls default positioning of children within the Stack.

Fit Properties

- StackFit.loose
- StackFit.expand
- StackFit.passthrough

Common Use Cases

- Profile image with online indicator
- Notification badges
- Image overlays
- Custom splash screens
- Floating buttons over images

5. ListView Widget

ListView displays scrollable content vertically or horizontally.

Types of ListView

1. ListView

Suitable for small lists.

2. ListView.builder

Creates items only when needed, improving performance.

Benefits:

- Faster rendering
- Lower memory usage
- Best for large datasets

3. ListView.separated

Allows separators between items.

Scrolling Behavior

Flutter provides smooth scrolling and customization options.

Custom Scroll Physics

Examples:

- BouncingScrollPhysics
- ClampingScrollPhysics
- NeverScrollableScrollPhysics

6. GridView Widget

GridView displays items in a grid format.

GridView.count

Creates a grid using a fixed number of columns.

GridView.builder

Efficiently creates large grids.

Grid Delegates

- SliverGridDelegateWithFixedCrossAxisCount
- SliverGridDelegateWithMaxCrossAxisExtent

Responsive Grids

Grid layouts automatically adjust based on screen size, making them ideal for tablets and phones.

Common Applications:

- Product catalogs
- Photo galleries
- Dashboard cards
- Menu screens

7. Responsive Design in Flutter

Responsive design ensures the UI looks good on different screen sizes.

Techniques

- MediaQuery
- LayoutBuilder
- Expanded and Flexible
- FractionallySizedBox

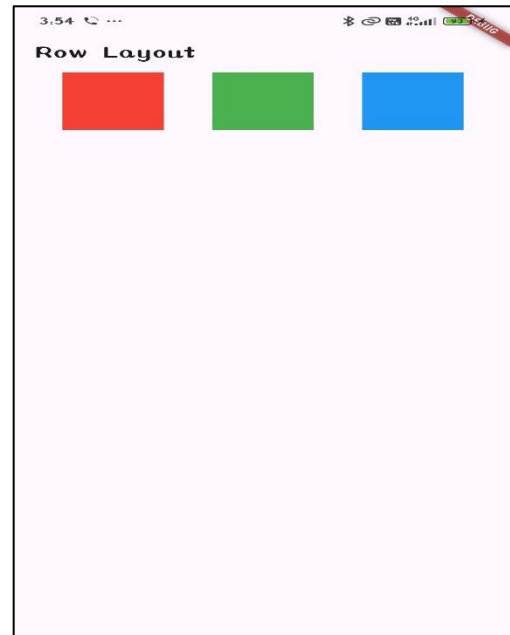
Benefits

- Better user experience
- Device compatibility
- Professional appearance

1. Row Layout :

```
import 'package:flutter/material.dart';

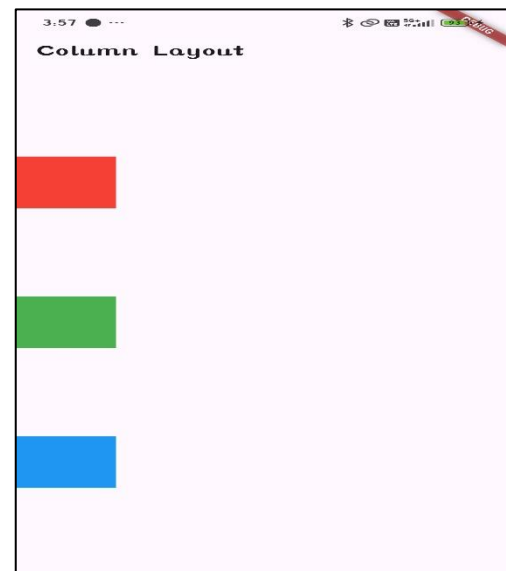
Run | Debug | Profile
void main() {
  runApp(MaterialApp(
    home: Scaffold(
      appBar: AppBar(title: Text("Row Layout")),
      body: Row(
        mainAxisAlignment: MainAxisAlignment.spaceEvenly,
        children: [
          Container(width: 80, height: 80, color: Colors.red),
          Container(width: 80, height: 80, color: Colors.green),
          Container(width: 80, height: 80, color: Colors.blue),
        ],
      ), // Row
    ), // Scaffold
  )); // MaterialApp
}
```



2. Column Layout :

```
import 'package:flutter/material.dart';

Run | Debug | Profile
void main() {
  runApp(MaterialApp(
    home: Scaffold(
      appBar: AppBar(title: Text("Column Layout")),
      body: Column(
        mainAxisAlignment: MainAxisAlignment.spaceEvenly,
        children: [
          Container(height: 80, width: 80, color: Colors.red),
          Container(height: 80, width: 80, color: Colors.green),
          Container(height: 80, width: 80, color: Colors.blue),
        ],
      ), // Column
    ), // Scaffold
  )); // MaterialApp
}
```



3. Expanded Layout

```
import 'package:flutter/material.dart';

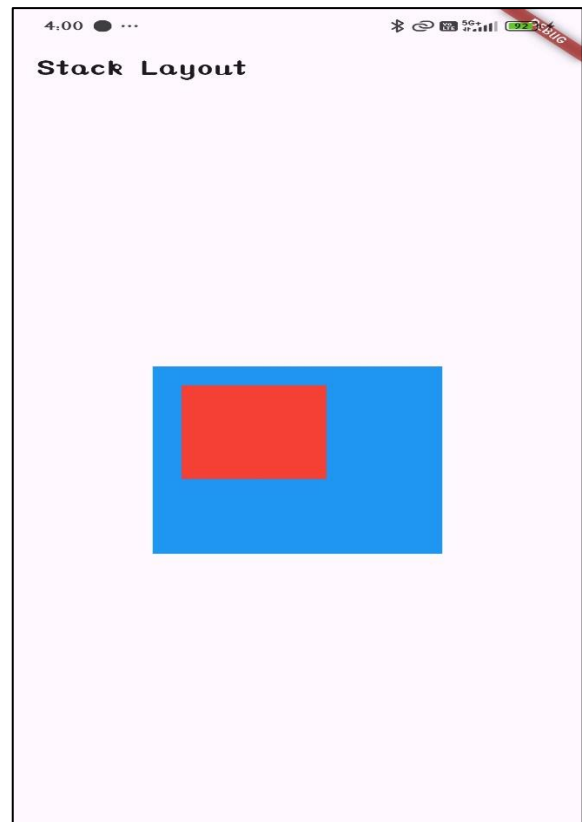
Run | Debug | Profile
void main() {
  runApp(MaterialApp(
    home: Scaffold(
      appBar: AppBar(title: Text("Expanded Widget")),
      body: Row(
        children: [
          Expanded(
            flex: 1,
            child: Container(color: Colors.red, height: 100),
          ), // Expanded
          Expanded(
            flex: 2,
            child: Container(color: Colors.green, height: 100),
          ), // Expanded
        ],
      ), // Row
    ), // Scaffold
  )); // MaterialApp
}
```



4. Stack Layout

```
import 'package:flutter/material.dart';

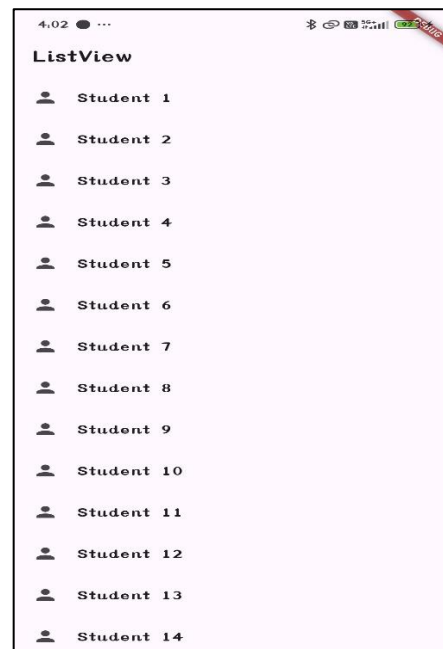
Run | Debug | Profile
void main() {
  runApp(MaterialApp(
    home: Scaffold(
      appBar: AppBar(title: Text("Stack Layout")),
      body: Center(
        child: Stack(
          children: [
            Container(
              width: 200,
              height: 200,
              color: Colors.blue,
            ), // Container
            Positioned(
              top: 20,
              left: 20,
              child: Container(
                width: 100,
                height: 100,
                color: Colors.red,
              ), // Container
            ), // Positioned
          ],
        ), // Stack
      ), // Center
    ), // Scaffold
  )); // MaterialApp
}
```



5. ListView Layout

```
import 'package:flutter/material.dart';

Run | Debug | Profile
void main() {
  runApp(MaterialApp(
    home: Scaffold(
      appBar: AppBar(title: Text("ListView")),
      body: ListView.builder(
        itemCount: 20,
        itemBuilder: (context, index) {
          return ListTile(
            leading: Icon(Icons.person),
            title: Text("Student ${index + 1}"),
          ); // ListTile
        },
      ), // ListView.builder
    ), // Scaffold
  )); // MaterialApp
}
```



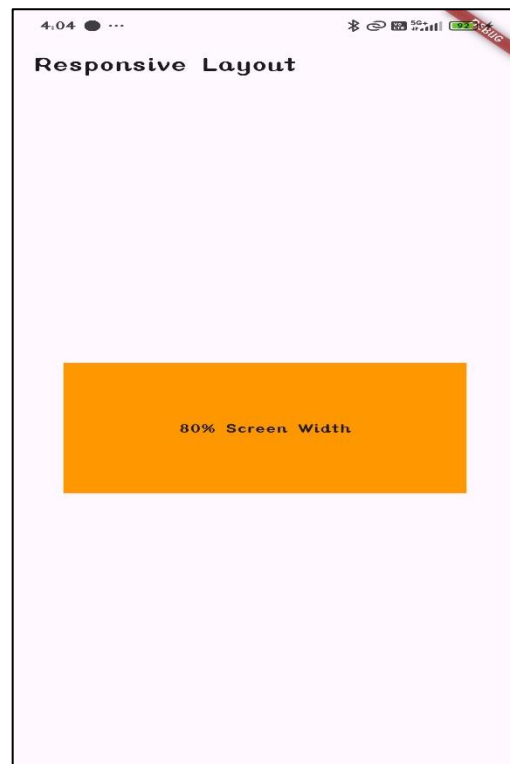
6. Responsive Layout

```
import 'package:flutter/material.dart';

Run | Debug | Profile
void main() {
  runApp(MaterialApp(
    home: ResponsiveDemo(),
  )); // MaterialApp
}

class ResponsiveDemo extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    double width = MediaQuery.of(context).size.width;

    return Scaffold(
      appBar: AppBar(title: Text("Responsive Layout")),
      body: Center(
        child: Container(
          width: width * 0.8,
          height: 150,
          color: Colors.orange,
          child: Center(
            child: Text("80% Screen Width"),
          ), // Center
        ), // Container
      ), // Center
    ); // Scaffold
  }
}
```



7. Complex Layout (Row , Column , Expanded)

```
import 'package:flutter/material.dart';

Run | Debug | Profile
void main() {
  runApp(MaterialApp(
    home: Scaffold(
      appBar: AppBar(title: Text("Complex Layout")),
      body: Padding(
        padding: EdgeInsets.all(10),
        child: Row(
          children: [
            CircleAvatar(radius: 40),
            SizedBox(width: 10),
            Expanded(
              child: Column(
                mainAxisAlignment: MainAxisAlignment.min,
                crossAxisAlignment: CrossAxisAlignment.start,
                children: [
                  Text(
                    "Sneha Tumaskar",
                    style: TextStyle(
                      fontSize: 18,
                      fontWeight: FontWeight.bold,
                    ), // TextStyle
                  ), // Text
                  Text("Flutter Developer"),
                  Text("Pune, Maharashtra"),
                ],
              ), // Column
            ), // Expanded
          ],
        ), // Row
      ), // Padding
    ), // Scaffold
  )); // MaterialApp
}
```

